

A Federated Learning Backdoor Attack Defense Method Based on Dual Attention Mechanism

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Abstract—As a distributed machine learning paradigm, federated learning allows clients to collaboratively train models without sharing their private data, effectively solving data privacy issues in edge computing scenarios. However, recent studies have shown that neural network models in federated learning are vulnerable to backdoor attacks, which make the global model give wrong inference results in a high-confidence manner, such as recognizing stop signs as speed limit signs in the image classification task. This will have serious consequences. Aiming at the problem that the existing federated learning defense methods take a long time to compute and cannot destroy the matching relationship between triggers and backdoors, a federated learning backdoor attack defense based on dual attention mechanism (FDDAM) is proposed. The model weights are dynamically adjusted during training process, no additional models are required, and the calculation time is shorter. First, in order for the model to ignore triggers, the enhancement on image semantics is performed and then build channel attention map. Second, in order to destroy the matching relationship between triggers and backdoors, a feature map space transformation network is constructed. Finally, in order to improve the defense success rate, the channel attention map and the spatial attention map are weighted to construct a dual attention network. Experiments with FDDAM on image classification datasets show an average increase of 1.68% and 3.11% in model accuracy and defense success rate, and an average reduction of 1.85 times in computation time compared to the benchmark method.

Keywords—Edge computing, Federated learning, Image classification, Backdoor attack, Dual attention

I. INTRODUCTION

With the rapid development of artificial intelligence, deep learning^[1] has shown obvious advantages in most fields. Deep learning has made breakthroughs in various industries such as transportation, medical, and manufacturing. Uploading large amounts of training data to the cloud will lead to high communication consumption. To solve the latency problem in cloud computing^[2], edge computing^[3] sinks computing nodes to the user side of the network edge to compute and process data at the edge of the network, effectively reducing the communication consumption between users and edge servers. While uploading private data to edge servers by users may lead to data privacy leakage problems. To address this challenge, federated learning has emerged as a highly promising solution. Participants do not need to upload their private data to the central server, but only need to train the model locally using private data and upload the model update

information to the central server. In data privacy-sensitive^{[7][8]} scenarios, federated learning has shown powerful advantages.

Despite the above advantages of federation learning, the security problem still exists because federation learning applications often contain a large number of deep neural networks. Several studies have shown^{[9][10]} that federation learning systems are vulnerable to backdoor attacks^[5]. On the one hand, during the client-side local training phase, an attacker can launch a backdoor attack by poisoning the client private dataset^[11] or hijacking the client node to send arbitrary gradients^[12]. On the other hand, while clients upload gradients through the communication channel, the attacker can maliciously tamper with the gradient information^{[13][14]}, leading to the injection of a backdoor into the global model. In the image classification task, how to build a federated learning system robust to backdoor attacks is still in its infancy. Defending against backdoor attacks in federated learning has become a new challenge. Therefore, it is of great theoretical and practical importance to study federated learning backdoor attack defense in image classification tasks.

In this paper, our main contributions are as follows:

- 1) By encoding the channel information, we construct a channel attention map and perform enhancement learning of image semantics, which makes the model ignore triggers. By constructing a spatial attention map, we destroy the matching relationship between the trigger and backdoor.
- 2) In order to improve the defense success rate, we construct a dual attention network with a weighted fusion of channel attention map and spatial attention map.

II. RELATED WORK

Gu et al.^[5] first proposed the powerful threat of backdoor attack. He gave a backdoor attack model BadNets. The goal of a backdoor attack is to infect a benign model so that it exhibits poor performance on some backdoor tasks specified by the attacker while maintaining high accuracy for the main task during the inference phase. Taking the image classification task as an example, the backdoor model maintains a high classification accuracy for clean input. However, when the input contains triggers, the backdoor model will make wrong classification behaviors. For example, in the field of autonomous driving, an infected backdoor model may recognize the stop sign as a speed limit sign, which will cause serious consequences.

In recent years, some scholars have proposed backdoor attack defense methods for image classification federated learning applications. Li et al.^[17] conducted a detailed study on static triggers. Based on this, they proposed a transformation-based backdoor attack defense method. The core idea is to transform the image and destroy the characteristics of the trigger through transformation. The advantage of this method is that the permission requirements for clean data are not high, and the computational cost is low. However, the limitation is that the transformation method needs to be pre-defined by humans, which is not universal. When faced with higher-strength attack algorithms, such as enhanced backdoor attacks that are robust to transformation, its performance is poor. Inspired by image style transfer^[19], Vasquez et al.^[18] proposed a backdoor attack defense method (Content-Focus, ConFoc) based on retraining. The key step is to generate a batch of "cure" dataset with the same content but different styles. The "cure" dataset is used to retrain the model, and the retrained model will pay more attention to the semantic content of the image, ignoring triggers. The core idea of this method is similar to the attention mechanism^{[20][21]}. For an image to be classified, the model should pay more attention to the content of the image, that is, the semantic information it represents, rather than the trigger images irrelevant to the semantic content, such as black and white pixel squares. However, the disadvantage of the ConFoc method is that the healing data set required for retraining depends on the original data set, and retraining also requires greater computational overhead. If in a federated learning application, each client is retrained locally during local training. Then the computational cost will become twice the original cost, and the convergence time of the global model will be greatly increased. At the same time, ConFoc utilizes the black-box nature of the neural network^[20], which does not have good interpretability.

This paper adopts a dual attention mechanism to dynamically adjust the model weights during the local training

phase of the federated learning client, without additional models or retraining processes. The computation time is lower than that of retraining defense methods. In addition, in order to destroy the matching relationship between triggers and backdoors and improve the defense success rate, the two attention maps are weighted and fused to construct a dual attention network. Compared with the existing backdoor attack defense methods in the image classification task, FDDAM reduces the computation time and improves the defense success rate.

III. METHOD

The FDDAM framework is shown in Fig. 1. FDDAM consists of five parts: input layer, semantic enhancement layer, feature map transformation layer, attention fusion layer, and output layer. The input data is the feature map in the hidden layer of the global model, and the output data is the feature map after dual attention is formed. The semantic enhancement layer is used to strengthen the model's learning of image semantics, compress and encode feature channel information, reduce the amount of parameters and computational costs. The pooled feature maps are fed into the neural network to learn the dependencies between feature channels and build a channel attention map. The feature map transformation layer uses a convolutional neural network to complete transformation parameter generation and coordinate mapping. Based on this mapping relationship, pixel sampling is performed and a spatial attention map is constructed to perform spatial transformation on the feature map. The attention fusion layer is used to perform weighted fusion of the channel attention map and the spatial attention map, and then generate the final dual attention map. This makes the model focus on image semantics, destroys the matching relationship between triggers and backdoors, and achieves the purpose of defending against backdoor attacks.

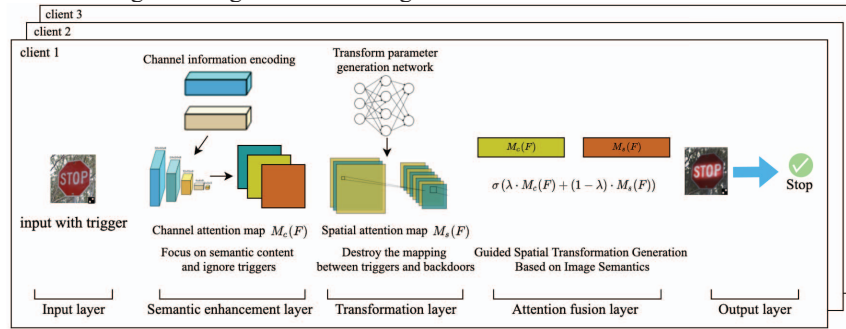


Fig. 1. FDDAM Frame diagram

A. Semantic Enhancement Learning

This section firstly compresses and encodes the channel information to enhance the semantic features of the image and reduce the size of the deep dimension feature map space, reducing the amount of parameters and computational cost. The channel feature vector is then fed into the neural network to build a channel attention map, which makes the model pay more attention to image semantics and ignore triggers.

The goal of image semantic enhancement learning is to make the model pay more attention to the semantic content of the image and ignore the triggers maliciously added by the attacker. In the convolutional neural network, the features

extracted by the deep network are often more abstract and have more semantic information. Therefore, the channel information is compressed and encoded to enhance its semantic features. The channel information encoding consists of a convolution layer and a global average pooling. First, the feature map of each channel is formed through a convolution kernel of each size. And then the spatial information of the feature map is aggregated through the pooling function, that is, the feature map is compressed by channel to obtain the numerical distribution information of each channel feature map. Compared with average pooling, the global average pooling method has a larger receptive field. The pooled feature map with more semantic information can be extracted

by using the global information. The global average pooling formula is as Equation 1.

$$F(u_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j) \quad (1)$$

where u_c is the c -th channel element of the input feature map.

After the above steps, the $1 \times 1 \times C$ channel feature maps obtained will be used to construct the channel attention map in the next step.

FDDAM constructs a new channel attention map based on the channel feature map generated by the channel information

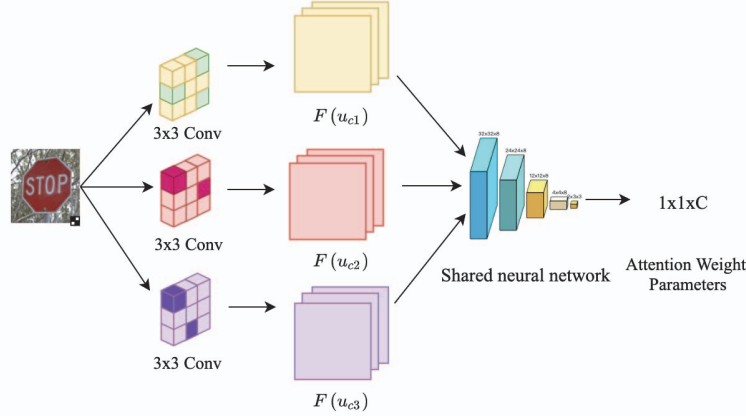


Fig.2. Attention mechanism

The channel attention map is obtained by multiplying the feature map weight value output by the neural network with the input feature map for scaling, as shown in Equation 2.

$$M_c(F) = F \cdot \sigma(\text{ReLU}(W_a^T h_k + b_k)) \quad (2)$$

where $F \in R^{H \times W \times C}$ is the input feature map, $M_c(F) \in R^{1 \times 1 \times C}$ is the output feature map, W_a^T is the trainable weight matrix, h_k is the k -th vector in the input feature map, b_k is the bias, and the dot product operator represents the product of a tensor of size $H \times W$ and a scalar of weights.

By constructing a channel attention map, the CNN network adaptively learns feature weights according to the loss function. By assigning weights, the feature maps that are effective for the main task are given more weight, while the feature maps that are invalid or less effective are assigned lower weights. The channel attention map is generated by the relationship between the feature channels. Each channel decomposed by the convolution kernel can be regarded as a feature extractor, and the importance of each feature channel is obtained by the network autonomous learning method. This allows the model to learn to pay attention to a certain part of a given input image. That is, by building a channel attention map, the model pays more attention to the subject to be recognized in the image, rather than paying attention to the backdoor trigger.

B. Feature Map Transformation Network

This section constructs the feature map space transformation network. First, the spatial transformation matrix is regressed through the convolutional neural network. And then the pixel coordinate mapping relationship is

encoding, as shown in Fig. 2. The channel feature maps are sent to a shared neural network, and the hidden layer order of the shared neural network is: Full Connect, ReLU, Full Connect, Sigmoid. The information of each feature map is fused through two fully connected layers. The weight of the feature map is learned from these fully connected layers and the nonlinear activation layer, and can be trained end-to-end. The parameter matrix W of the neural network is used to model correlation between feature channels. Finally, a value is obtained through the Sigmoid gate function, that is, the actual weight of each channel feature map.

calculated with the input feature map. Finally, the pixels are sampled using the linear difference to form a spatial attention map, which destroys the matching relationship between the trigger and the backdoor.

The transformation parameter generation network consists of several convolutional layers and fully connected layers. The last layer is a regression layer. The input of the parameter generation network is a tensor, and the output is the transformation parameter θ . The dimension of θ depends on the type of transformation learned by the network, such as affine transformation $\theta \in R^{2 \times 3}$. The following is an example of affine transformation. The initial value of θ is set to the identity transformation matrix, and the parameter θ are continuously updated through the loss function. After the parameters are obtained, the transformation process between the input and the output cannot be directly described. The coordinate mapping relationship between the input and the output needs to be established, that is, the index relationship between the coordinates is determined in the normalized grid. For example, θ describes the identity transformation, then the coordinate relationship between input and output is one-to-one correspondence.

For affine transformation, the coordinate mapping relationship between input and output is defined as follows.

$$\begin{pmatrix} x'_i \\ y'_i \end{pmatrix} = T_\theta(G_i) = A_\theta \begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix} = \begin{bmatrix} \theta_{11} & \theta_{12} & \theta_{13} \\ \theta_{21} & \theta_{22} & \theta_{23} \end{bmatrix} \begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix} \quad (3)$$

where s represents the coordinate point of the input feature tensor, A_θ is the transformation matrix learned by the

parameter generation network, and t represents the coordinate point of the output feature tensor.

After the transformation parameter and coordinate mapping are generated, the corresponding coordinate point of each coordinate point of the transformed output feature tensor to the input feature tensor can be determined. The next step is to extract the corresponding coordinate pixel value in the input feature tensor. When the coordinate value is an integer, the pixel value can be directly extracted. However, the coordinate point corresponding to the input tensor may be a floating number, and the pixel sampling part needs to be a differentiable step, allowing the gradient to be back-propagated. Therefore, bilinear interpolation is used in this paper to calculate the pixel value of the corresponding point, as shown in Equation 4.

$$V_i^c = \sum_n \sum_m U_{nm}^c \max(0, 1 - |x_i^s - m|) \max(0, 1 - |y_i^s - n|) \quad (4)$$

where $\max$ is a differentiable function that selects the pixel position where the distance between the output and the coordinate point (x_i^s, y_i^s) is less than 1, and the point with the closer distance is assigned a higher weight.

C. Dual Attention Network

In this section, the channel attention map and the spatial attention map are weighted and fused to construct a dual attention network.

The channel attention map makes the model pay more attention to the semantic information of the image, but essentially the relevant features of the trigger are still retained in the hidden layer feature map. While the spatial attention map solves this problem well, destroying the location and appearance features of the trigger. At the same time, the channel attention map is guided by image semantics and assists the generation of transformation parameters in the feature map fusion network. The dual attention weighted fusion formula is shown in Equation 5.

$$Dual_Atten = \sigma(\lambda \cdot M_s(F) + (1 - \lambda) \cdot M_c(F)) \quad (5)$$

where $M_c(F)$ is the channel attention map, $M_s(F)$ is the spatial attention map, σ is the sigmoid function, and λ is the weight hyperparameter that controls the fusion ratio.

IV. EXPERIMENTS

A. Experiment Setup

a) Dataset and Model

There are four datasets chosen for the experiments, which are CIFAR-10, GTSRB, B TSR, and VGGFace2 dataset. CIFAR-10 is a commonly used natural scene recognition dataset, with image size of 32*32*3, training dataset size of 50,000, and test dataset size of 10,000. GTSRB and B TSR are both traffic sign recognition datasets, including stop signs, speed limit signs, etc. Traffic sign recognition datasets are commonly used in the field of autonomous driving, which has very high security requirements, and therefore are often used in backdoor attack defense experiments. The GTSRB dataset has 43 categories, with 34,799 training images and 12,630 test images. VGGFace2 is a face recognition dataset with about 50,000 training images and 10,000 test images, which contains more face categories compared to VGGFace. As shown in TABLE I. :

TABLE I. OVERVIEW OF DATASET

Dataset	Image size	Training set	Test set	Label
CIFAR-10	32*32*3	50000	10000	10
GTSRB	32*32*3	34799	12630	43
B TSR	224*224*3	4500	2500	62
VGGFACE2	224*224*3	50000	10000	162

b) Attack Strategy

The attack algorithm used in the experiment is the classic backdoor attack algorithm BadNets proposed by Gu et al.^[5], which is the same as the attack algorithm used in the benchmark method selected in this paper. BadNets are representative of backdoor attack algorithms for training data poisoning. For each dataset, this paper randomly selects a label as the target label maliciously set by the attacker, embeds the trigger into a part of the training data and modifies the label of the part of the toxic data to the target label. The trigger style is a black and white pixel image, and the position is in the lower right corner of the image. The trigger pattern size is 3*3 for a dataset with an image size of 32*32, and 18*18 for a dataset with an image size of 224*224.

c) Benchmark Method

The benchmark methods for experimental comparison are as follows:

(1) Standard Training: Standard training is to use clean data for training, and its purpose is to compare the accuracy of benign models.

(2) Februus^[15]: An additional GAN model is used to process the input image.

(3) ConFoc^[18]: Use a batch of images with the same content but different styles to retrain the model to focus more on the semantic information of the images in order to defend against backdoor attacks.

(4) Transformation-Based Defense (TBD)^[17]: Using a preprocessing module to spatially transform the input image and defending against backdoor attacks by changing the appearance or location of triggers.

d) Evaluation

(1) Accuracy: The classification accuracy intuitively describes the training effect of the model. The higher the accuracy, the better the model training effect.

(2) Defense Success Rate: The defense success rate indicates the effect of defending against backdoor attacks. The higher the defense success rate, the worse the backdoor attack effect. The defense success rate formula is expressed as formula 6:

$$DSR(F^p, D) = 1 - \frac{D^{target}}{D^{poison}} \quad (6)$$

where F^p is the infected backdoor model, D^{poison} is the total number of samples with triggers, and D^{target} is the number of samples that the model predicts as the target label.

(3) Convergence time: The time required for local training of the client in each round. The faster the convergence time, the less computing resources are consumed.

B. Analysis of Results

a) Defense Success Rate Comparative Analysis

From TABLE II. , we can see the powerful threat of backdoor attacks. Applying the BadNets attack to the standard training process can achieve an attack success rate of approximately 100%, that is, the defense success rate is approximately 0%. After using ConFoc for defense, the defense success rate increased significantly on each dataset, rising to 99.5% on the CIFAR-10 dataset, rising to 99.4% and 99.5% on the GTSRB and BTSR respectively and rising to 99.2% on the VGGFace2 dataset. Februous and TBD are

similar, the defense success rate after applying Februous rise to 99.3%, 99.0%, 99.2% and 99.3% respectively. The defense success rate after applying TBD rise to 98.4%, 98.2%, 98.3% and 98.6% respectively. After applying the defense method based on the dual attention mechanism, the average defense success rate increased to about 99.5%, which shows that the defense method based on the dual attention mechanism performed better in terms of defense effect.

TABLE II. DEFENSE SUCCESS RATE COMPARISON EXPERIMENT

Model	Resnet34			
Attack Algorithm	BadNets			
DataSet	CIFAR-10	GTSRB	BTSR	VGGFace2
Method				
Standard Training	0.0	0.1	0.0	0.0
ConFoc	96.5	96.4	96.5	96.2
TBD	95.4	95.2	95.3	95.6
Februous	96.3	96.0	96.2	96.3
FDDAM (Ours)	98.9	99.1	99.3	98.8

Compared with the benchmark methods, the FDDAM proposed in this paper has better defense performance, and the defense success rate is increased by 3.11% on average compared to the benchmark methods. This is because FDDAM makes the model ignore the trigger by constructing the channel attention map and the spatial attention map, destroying the mapping relationship between trigger and backdoor. At the same time, a dual attention network is constructed to fuse the channel attention map and the spatial attention map by weight to improve the defense success rate.

b) Computational Time Comparative Analysis

Lower computation time means less computational resource requirements and faster convergence time. The experimental environment is Ubuntu16.04, Python 3.7.5, and Pytorch 1.3.1. The single running time of the algorithm is shown in TABLE III. . The calculation time is obtained by taking the average of ten independent experimental results for each algorithm.

TABLE III. COMPUTATION TIME COMPARISON EXPERIMENT (S)

Model	Resnet34			
Attack Algorithm	BadNets			
DataSet	CIFAR-10	GTSRB	BTSR	VGGFace2
Method				
ConFoc	~10000	~7000	~8000	~9000
TBD	~5000	~5500	~5000	~6000
Februous	~7000	~6000	~7000	~7500
FDDAM (Ours)	~3500	~4000	~3500	~4000

It can be seen from TABLE III. that the calculation of ConFoc based on retraining takes the most time, and the average time on different datasets is about 8500 seconds. This is because retraining not only needs to generate a batch of healing datasets, but also needs to perform backpropagation. The transformation-based defense method TBD only involves an additional simple spatial transformation in the calculation process, so the calculation time is the least, and the average time is only about 5000 seconds. Although the calculation cost is the least, the defense performance of TBD as a cost is not good. That is because the spatial transformation method needs to be pre-defined by humans, and it performs poorly in the face of

backdoor attacks that are robust to transformations. The calculation time of the method in this paper is lower than that of the benchmark method, and the average time is about 3500 seconds on different data sets. This is because FDDAM dynamically adjusts the weights of neurons in the forward and backward propagation of network training, and does not require attaching additional models or retraining process.

c) Dual Attention Fusion Ratio Analysis

The dual attention network is the core part of FDDAM, so setting an appropriate fusion ratio λ is an important task.

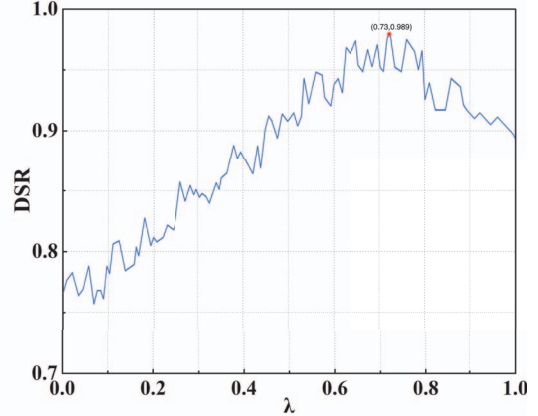


Fig.3 Trend of DSR of FDDAM with the value of λ

In this section, through observing the change trend of the defense success rate with λ , we select the highest point of DSR value as the final value of λ . The change trend of FDDAM defense success rate with the fusion ratio λ is shown in Fig. 3. It can be seen from the figure that the DSR curve as a whole shows a trend of rising first and then falling. The highest point value is 0.73, and the corresponding DSR value is 0.98. It can be seen that building the spatial attention map is the key to defense, and the channel attention map can play an auxiliary role.

V. CONCLUSIONS

This paper comprehensively considers the mechanism of backdoor attack and the characteristics of the trigger itself,

and proposes a federated learning backdoor attack defense method based on the dual attention mechanism. FDDAM first encodes the channel information, constructs the channel attention map. And then perform spatial transformation on the feature map to destroy the trigger-backdoor matching relationship. Finally, the two attention maps are weighted and fused to improve the defense success rate. Experiments on image classification datasets confirm the effectiveness of the method. The limitation of the existing work is that we only consider image classification tasks at present, and further research can be done on speech and video data in the future.

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